

ISHAN KUMAR SAHU

B.Tech CSE (Data Science), Bhilai Institute of Technology, Durg
ishanksahu@bitdurg.ac.in | (+91) 9302597193 | [LinkedIn](#) | [GitHub](#) | [HuggingFace](#)

INTRODUCTION

Third-year B.Tech CSE (Data Science) student at Bhilai Institute of Technology, Durg (CPI: 8.31). Previous work spans low-resource NLP, edge-deployed sign language recognition, and IoT-based civic infrastructure. **Smart India Hackathon 2025 winner** (Hardware Edition) and recipient of **₹2 Lakh government research funding** under the DST i-NIDHI scheme for real-world municipal deployment of BinSense.

RESEARCH INTERESTS

Deepfake detection and synthetic media forensics · Adversarial ML and generative model robustness ·

Low-resource NLP and speech processing · Computer vision for autonomous systems · Edge-deployed AI under real-world hardware constraints

RESEARCH & PROFESSIONAL EXPERIENCE

ML / DL Intern — MIST Lab, IIT Bhilai

June 2026 – Sept 2026

Supervisor: Dr. SK Subidh Ali | Onsite, Durg, Chhattisgarh

- Researching **generator-agnostic deepfake detection** — building models that identify AI-generated facial images regardless of the source generator or architecture.
- Implementing and extending **GAKer** (adversarial example generation) and **conditional diffusion models**; studying cross-generator generalization under distribution shift.
- Applying deep learning and image processing techniques toward a unified detection framework robust to unseen generators.

ML / LLM Intern — Spaceborn Autonomous Systems

April 2026 – Present

Remote | Autonomous drone AI — computer vision and language model track

- Building CV features for autonomous drone navigation: **gesture control, path and trajectory prediction**, and obstacle avoidance using SLAM sensor data.
- Contributing to the LLM pipeline via **LoRA and QLoRA fine-tuning**; handling output customization, training automation, and dataset generation and integration.

PROJECTS

ShrutiAI — Offline sign language translation system

Tech stack: Raspberry Pi, MediaPipe, TensorFlow/Keras, React, Node.js [\[Video\]](#) [\[Site\]](#)

- Edge-deployed ISL gesture recognition on Raspberry Pi — fully offline, no internet dependency — converting hand gestures to English text in real time via MediaPipe landmark extraction.
- Dense neural network trained on a self-curated dataset of **1,38,216 datapoints across 70 ISL signs**; achieved **99.89% test accuracy** (loss: 0.0037) with 2ms per-sample inference on the trained model.
- Won Smart India Hackathon 2025 — Hardware Edition** (PSID 25247). React + Node.js interface renders translations with sub-second display refresh.

BinSense — IoT-based smart waste management platform

Tech stack: ESP32, ThingSpeak, Firebase, React, Django [\[GitHub\]](#) [\[Site\]](#) [\[Demo\]](#)

- IoT sensor network across **10 deployed dustbin units** streaming fill-level telemetry to Firebase; automated threshold-based alerts to municipal authorities.

- Waste pattern analytics module estimating **65% reduction in redundant collection trips**. Received **₹2 Lakh government research funding** under DST i-NIDHI scheme, CSVTU.
- **1st Prize, Code of the Phoenix Hackathon** — IIIT Naya Raipur, 2024.

AksharDhara — Low-resource dialect speech translation pipeline

Tech stack: OpenAI Whisper, Meta NLLB-200, PyTorch, Hugging Face [\[GitHub\]](#) [\[HuggingFace\]](#)

- End-to-end speech-to-text translation pipeline for Chhattisgarhi — a low-resource dialect spoken by **~18 million people** with virtually no prior NLP tooling.
- Fine-tuned Whisper (ASR) and NLLB-200-600M (translation) on a **self-curated dataset of 20,000 sentence pairs**; improved chrF score by **46.34%** over base model baseline, with ~1s end-to-end latency per utterance.

TECHNICAL SKILLS

Languages: Python, JavaScript

ML / Computer Vision: PyTorch, TensorFlow, CNNs, MediaPipe, Ultralytics YOLO, scikit-learn

LLMs / NLP: LoRA, QLoRA, PEFT, Hugging Face Transformers, OpenAI Whisper, NLLB-200

Web & Backend: React, Node.js, Django, Docker, Tailwind CSS, Bootstrap, REST APIs

Embedded / IoT: Raspberry Pi, ESP32/ESP8266, ThingSpeak, Firebase, Sensors

ACHIEVEMENTS & HONOURS

- **Winner, Smart India Hackathon 2025 — Hardware Edition** (PSID 25247). National-level government-run competition; developed ShrutiAI.
- **₹2 Lakh Research Grant** — CSVTU under DST i-NIDHI scheme, Govt. of India. Awarded for BinSense (2025).
- **National Finalist, HULT Prize 2026** — Pitched BinSense at IIT Bombay at the India National Finals.
- **1st Prize, Code of the Phoenix Hackathon** — IIIT Naya Raipur, 2024. Open category.
- **Runner Up, BIT-SHINE Hackathon** — Departmental level, BIT Durg, 2023.
- **1st Prize, Disaster Idea-thon** — IEEE BITD, 2025.

EDUCATION

Bhilai Institute of Technology, Durg

Sept 2023 – Sept 2027

B.Tech, Computer Science & Engineering (Data Science) **CPI: 8.31** (till 5th semester)

Relevant coursework: Data Structures & Algorithms, Machine Learning, DBMS, Computer Networks, Statistics & Probability

Holy Cross Senior Secondary School, Raipur

2022

XII CBSE — 75%

Holy Cross Senior Secondary School, Raipur

2020

X CBSE — 85.7%

POSITIONS OF RESPONSIBILITY

Vice Secretary — The ASTRO Club, BIT Durg

2024 – Present

Astronomy and astrophysics student society. Organised stargazing events and outreach activities.

Secretary — Computer Science Engineering Association (CSEA), BIT Durg

2024 – Present

Departmental technical society. Coordinated workshops, coding competitions, and technical events.

Events & Logistics Lead — TEDxBIT Durg

2024

Core organising committee. Managed end-to-end logistics, speaker coordination, and on-ground operations.

COURSEWORK & TRAINING

- **CS50x** — Introduction to Computer Science, Harvard University (via OpenCourseWare). Completed.